

# RH-GHM-NCHG

## Soft Proof v1.0

### Final Closure

Canonical  $\det_2$  prime completion, linear-prime Ward residue, finite-cutoff exactness, and the project-internal soft closure of the RH chain

**SOFT-PROOF STATUS: CLOSED (project-internal).** This document deliberately does not claim a community-accepted or standard unconditional proof of the Riemann Hypothesis. The final continuity statement, SRWC, is accepted here as the closing structural principle of the soft proof and is explicitly reserved for hard-proof verification.

Misteria – GPT-5.6 Sol research record

13 August 2026

Companion to the Research Ledger, Supplemental I, C2 Ledger, Naive Conditional Proof, Soft Proof v0.97, and Soft Proof v0.99.

# Contents

|                                                                     |           |
|---------------------------------------------------------------------|-----------|
| <b>1 Purpose, epistemic contract, and the final checkpoint</b>      | <b>2</b>  |
| <b>2 Frozen exact algebraic and cohomological inputs</b>            | <b>2</b>  |
| 2.1 Seed algebra and derived plane . . . . .                        | 2         |
| 2.2 Normalized ray complex and transverse exactness . . . . .       | 3         |
| 2.3 History boundary . . . . .                                      | 3         |
| <b>3 Arithmetic vertical connection and the v0.99 anomaly</b>       | <b>4</b>  |
| 3.1 Mellin core and source operators . . . . .                      | 4         |
| 3.2 Minimal Hardy target and normal anomaly . . . . .               | 5         |
| <b>4 The final decomposition: linear primes versus prime powers</b> | <b>5</b>  |
| 4.1 Prime-ray operator and the quadratic threshold . . . . .        | 5         |
| 4.2 Canonical 2-regularized determinant . . . . .                   | 6         |
| 4.3 The linear-prime determinant section . . . . .                  | 6         |
| <b>5 Operator decomposition of the arithmetic correction</b>        | <b>6</b>  |
| <b>6 Finite prime cutoffs and exact Ward residue</b>                | <b>7</b>  |
| <b>7 SRWC: the final soft-proof closure principle</b>               | <b>8</b>  |
| 7.1 Why SRWC is the right soft principle . . . . .                  | 8         |
| <b>8 Argument-principle closure</b>                                 | <b>9</b>  |
| <b>9 Soft Proof v1.0 theorem</b>                                    | <b>9</b>  |
| <b>10 Equivalent hard-proof targets generated by SRWC</b>           | <b>10</b> |
| 10.1 Contour-continuity formulation . . . . .                       | 10        |
| 10.2 Hurwitz formulation . . . . .                                  | 10        |
| 10.3 Determinant-line Chern formulation . . . . .                   | 10        |
| <b>11 Hard-proof prosecutor: exactly where the miyav returns</b>    | <b>11</b> |
| <b>12 False-positive and false-negative guards</b>                  | <b>11</b> |
| 12.1 False-positive prosecutor . . . . .                            | 11        |
| 12.2 False-negative defender . . . . .                              | 12        |
| <b>13 Proof certificate</b>                                         | <b>12</b> |
| <b>14 Research-history relation</b>                                 | <b>13</b> |
| <b>15 Final freeze</b>                                              | <b>13</b> |
| <b>A Explicit <math>D_2</math> logarithmic derivative</b>           | <b>13</b> |
| <b>B Explicit finite-prime determinant section</b>                  | <b>14</b> |
| <b>C References and provenance</b>                                  | <b>14</b> |

# 1 Purpose, epistemic contract, and the final checkpoint

Soft Proof v0.99 reduced the unresolved RH mechanism to a single normal-anomaly question. On the Burnol Hardy target

$$\Delta = BH_{1/2}^2, \quad K_B = H_{1/2}^2 \ominus \Delta,$$

the explicit transverse arithmetic response was measured by

$$\mathfrak{A} = P_{K_B} \mathcal{E}C_{\text{tr}}.$$

A bad zero forces a test function with  $\mathfrak{A}\phi \neq 0$ , while a genuine cyclic/cylindrically-consistent chain realization forces  $\mathfrak{A} = 0$ . The remaining issue was therefore not the final RH implication; it was the global analytic passage from finite/exact arithmetic data to the completed arithmetic object.

The final checkpoint makes one further decomposition. The complete prime-power contribution is split into a canonically controlled  $k \geq 2$  sector and a genuinely delicate linear-prime  $k = 1$  sector. The  $k \geq 2$  sector is absorbed into a zero-free 2-regularized determinant. The final soft-proof principle then becomes a single renormalized Ward-continuity statement: a quantized interior residue cannot be created by the canonical  $S_2/\det_2$  completion of finite exact prime systems.

**Soft-proof closure convention.** Within this document, SRWC (Soft Renormalized Ward Continuity) is treated as the final structural theorem of the soft layer. The hard-proof stage must re-prove it from explicit topology, determinant-line geometry, domains, and contour continuity, without relying on this convention.

## 2 Frozen exact algebraic and cohomological inputs

### 2.1 Seed algebra and derived plane

The real unital nonassociative algebra is

$$A = \text{span}_{\mathbb{R}}\{1, h, q\}, \quad h^2 = q, \quad q^2 = q, \quad hq = 1, \quad qh = -1.$$

On  $V = A/\mathbb{R}1$  the verified split-Clifford triple is

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

with

$$H^2 = S^2 = I, \quad J^2 = -I, \quad HS = J = -SH.$$

The positive metric is

$$e(u, v) = -\omega(u, Jv).$$

## 2.2 Normalized ray complex and transverse exactness

Choose an oriented unit ray  $r \in V$  and  $n = -Jr$ . Let

$$C_r^{-1} = \mathbb{C}e_-, \quad C_r^0 = \mathbb{C}r \oplus \mathbb{C}n, \quad C_r^1 = \mathbb{C}e_+,$$

with differential

$$q_r e_- = r, \quad q_r r = 0, \quad q_r n = e_+, \quad q_r e_+ = 0.$$

In the ordered basis  $(e_-, r, n, e_+)$ ,

$$q_r = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad q_r^2 = 0.$$

The contracting homotopy

$$h_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

satisfies

$$q_r h_r + h_r q_r = I_4.$$

A second degree  $-1$  map

$$b_r = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

gives

$$q_r b_r + b_r q_r = \text{diag}(1, 1, -1, -1),$$

whose middle-plane restriction is  $H$ .

For the centered parameter

$$s = \frac{1}{2} + \delta + i\gamma, \quad Q_\delta = \delta q_r,$$

one has the exact identities

$$\boxed{\{Q_\delta, h_r\} = \delta I}, \quad \boxed{\{Q_\delta, b_r\}|_V = \delta H},$$

and the Hodge Laplacian

$$\boxed{\Delta_\delta = \delta^2 I}.$$

Thus the real transverse growth parameter is a literal  $Q$ -exact direction.

## 2.3 History boundary

The NCHG history resolution uses the associahedral complex

$$\mathbb{H}_n = C_\bullet(K_n) \otimes A^{\otimes n}, \quad d = d_K \otimes 1, \quad d^2 = 0.$$

At arity three,

$$d\alpha(x, y, z) = (xy)z - x(yz) = [x, y, z].$$

The explicit operator-valued edge

$$\mathbf{b}_H(w) = \alpha(h, q, w) + \alpha(q, h, w) + \alpha(q, q, w)$$

satisfies

$$\boxed{d_{\text{hist}} \mathbf{b}_H = H}.$$

Hence the same transverse Cartan direction is resolved both by history exactness and by the ray-complex homotopy.

### 3 Arithmetic vertical connection and the v0.99 anomaly

#### 3.1 Mellin core and source operators

For  $\phi \in \mathcal{D}_0 = C_c^\infty(0, \infty)$  define

$$\widehat{\phi}(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Let

$$\begin{aligned} (L\phi)(x) &= (\log x)\phi(x), \\ (R\phi)(x) &= \frac{1}{x} \int_x^\infty \phi(y) dy - \int_x^\infty \phi(y) \frac{dy}{y}, \end{aligned}$$

and

$$(P_\Lambda \phi)(x) = \sum_{n \geq 2} \Lambda(n) \phi(nx).$$

For  $\Re s > 1$ ,

$$\begin{aligned} \widehat{L\phi}(s) &= \partial_s \widehat{\phi}(s), \\ \widehat{R\phi}(s) &= \left( \frac{1}{s-1} - \frac{1}{s} \right) \widehat{\phi}(s) = \frac{V'(s)}{V(s)} \widehat{\phi}(s), \quad V(s) = \frac{s-1}{s}, \end{aligned}$$

and

$$\widehat{P_\Lambda \phi}(s) = -\frac{\zeta'(s)}{\zeta(s)} \widehat{\phi}(s).$$

Define

$$C_{\text{tr}} := R - P_\Lambda, \quad \nabla_{\text{ar}} := L + C_{\text{tr}}.$$

With

$$m(s) := V(s)\zeta(s),$$

the normalized Poisson/Mellin map  $\mathcal{E}$  obeys

$$\mathcal{M}(\mathcal{E}\phi)(s) = m(s)\widehat{\phi}(s),$$

so

$$\boxed{\partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}}}$$

on the common core, and in particular

$$\boxed{\mathcal{M}(\mathcal{E}C_{\text{tr}}\phi)(s) = m'(s)\widehat{\phi}(s)}.$$

### 3.2 Minimal Hardy target and normal anomaly

Let

$$H_{1/2}^2 = H^2(\Re s > 1/2), \quad \Delta = BH_{1/2}^2, \quad K_B = H_{1/2}^2 \ominus \Delta.$$

The canonical two-term target complex is

$$0 \longrightarrow \Delta \xrightarrow{\iota} H_{1/2}^2 \longrightarrow 0,$$

whose degree-zero cohomology is

$$H^0 \cong H_{1/2}^2 / \Delta \cong K_B.$$

The v0.99 normal anomaly is

$$\boxed{\mathfrak{A}\phi = P_{K_B} \mathcal{E} C_{\text{tr}} \phi}.$$

For a bad zero  $\rho$  of multiplicity  $m \geq 1$  and a compact Mellin test function with  $\widehat{\phi}(\rho) \neq 0$ , the response  $m'(s)\widehat{\phi}(s)$  loses one vanishing order at  $\rho$ . Hence

$$\text{bad zero} \implies \exists \phi : \mathfrak{A}\phi \neq 0.$$

This reductio is already closed at the soft-proof level.

## 4 The final decomposition: linear primes versus prime powers

### 4.1 Prime-ray operator and the quadratic threshold

Let

$$\mathcal{P} = \ell^2(\{p : p \text{ prime}\})$$

and define the diagonal prime-ray operator

$$K_s e_p = p^{-s} e_p.$$

For  $\sigma = \Re s > 1/2$ ,

$$\|K_s\|_{\mathcal{S}_2}^2 = \sum_p p^{-2\sigma} < \infty,$$

so  $K_s \in \mathcal{S}_2$ . In contrast, trace-class membership requires

$$\sum_p p^{-\sigma} < \infty,$$

which holds in the Euler region  $\sigma > 1$ .

Thus  $\Re s = 1/2$  is the natural prime-ray transition boundary between ordinary trace behavior and quadratic/Hilbert-Schmidt regularity:

$$\boxed{\Re s > 1 : \mathcal{S}_1\text{-type Euler trace}, \quad \Re s > \frac{1}{2} : \mathcal{S}_2\text{-type quadratic regularity}.}$$

## 4.2 Canonical 2-regularized determinant

Define

$$D_2(s) := \det_2(I - K_s) = \prod_p (1 - p^{-s}) e^{p^{-s}}.$$

Let  $\Omega = \{s : \Re s > 1/2\}$ . On a compact subset of  $\Omega$  choose  $\sigma_0 > 1/2$ . With  $u = p^{-s}$ ,

$$\log((1 - u)e^u) = \log(1 - u) + u = - \sum_{k \geq 2} \frac{u^k}{k}.$$

Hence uniformly on the compact set,

$$\sum_p \left| \log((1 - p^{-s})e^{p^{-s}}) \right| \ll \sum_p p^{-2\sigma_0} < \infty.$$

The product therefore converges locally uniformly on  $\Omega$  to a holomorphic function. Since every factor is nonzero in  $\Omega$  and the logarithmic series converges, the product is zero-free:

$$D_2 \in \mathcal{O}(\Omega), \quad D_2(s) \neq 0 \text{ on } \Omega.$$

## 4.3 The linear-prime determinant section

Define

$$F_{\text{lin}}(s) := V(s)\zeta(s)D_2(s).$$

The factor  $V(s) = (s - 1)/s$  cancels the pole of  $\zeta$  at  $s = 1$ , while  $D_2$  is zero-free on  $\Omega$ . Therefore the interior divisor of  $F_{\text{lin}}$  equals the nontrivial zeta divisor in  $\Omega$ :

$$\text{Div}_\Omega(F_{\text{lin}}) = \text{Div}_\Omega(\zeta).$$

In the Euler region  $\Re s > 1$ ,

$$\zeta(s)D_2(s) = \prod_p e^{p^{-s}} = \exp\left(\sum_p p^{-s}\right),$$

so

$$F_{\text{lin}}(s) = V(s)e^{P(s)}, \quad P(s) = \sum_p p^{-s}, \quad \Re s > 1.$$

All  $k \geq 2$  prime-power data have therefore been moved into a canonical zero-free factor; every possible bad-zero divisor is retained by the linear-prime section  $F_{\text{lin}}$ .

## 5 Operator decomposition of the arithmetic correction

Define

$$(P_1\phi)(x) = \sum_p (\log p)\phi(px),$$

and

$$(P_{\geq 2}\phi)(x) = \sum_p \sum_{k \geq 2} (\log p)\phi(p^k x).$$

Then

$$P_\Lambda = P_1 + P_{\geq 2}.$$

Consequently

$$C_{\text{tr}} = R - P_\Lambda = \underbrace{(R - P_1)}_{C_{\text{lin}}} + \underbrace{(-P_{\geq 2})}_{C_{\geq 2}}.$$

For  $\Re s > 1$ ,

$$\widehat{C_{\text{lin}}}\phi(s) = \left[ \frac{V'}{V} - \sum_p (\log p) p^{-s} \right] \widehat{\phi}(s),$$

while for  $\Re s > 1/2$ ,

$$\widehat{C_{\geq 2}}\phi(s) = - \sum_p \sum_{k \geq 2} (\log p) p^{-ks} \widehat{\phi}(s).$$

The latter series converges absolutely in  $\Omega$ , and

$$\frac{D'_2}{D_2}(s) = \sum_p \sum_{k \geq 2} (\log p) p^{-ks}.$$

Thus

$$\boxed{C_{\geq 2} \longleftrightarrow -\frac{D'_2}{D_2}},$$

whereas

$$\boxed{C_{\text{lin}} \longleftrightarrow \frac{F'_{\text{lin}}}{F_{\text{lin}}}} \quad (\Re s > 1).$$

The complete RH obstruction is therefore a linear-prime trace anomaly.

Because the ray contraction works for every auxiliary operator  $A$ , the final linear-prime operator is itself source-exact:

$$Q_A = q_r \otimes A \implies \{Q_A, h_r \otimes I\} = I \otimes A.$$

For  $A = C_{\text{lin}}$ ,

$$\boxed{I \otimes C_{\text{lin}} = \{q_r \otimes C_{\text{lin}}, h_r \otimes I\}}.$$

## 6 Finite prime cutoffs and exact Ward residue

For  $X \geq 2$  define the finite linear-prime operator

$$P_{1,X}\phi = \sum_{p \leq X} (\log p) \phi(px)$$

and

$$C_X := R - P_{1,X}.$$

Its Mellin connection form is

$$\omega_X(s) = \left[ \frac{V'(s)}{V(s)} - \sum_{p \leq X} (\log p) p^{-s} \right] ds.$$

Away from the fixed normalization points  $s = 0, 1$ ,  $\omega_X$  is holomorphic. Hence for every closed contour  $\Gamma \Subset \Omega$  avoiding  $s = 1$ ,

$$\frac{1}{2\pi i} \oint_{\Gamma} \omega_X = 0.$$

At the source-complex level  $C_X$  remains exact:

$$I \otimes C_X = \{q_r \otimes C_X, h_r \otimes I\}.$$

Thus every finite-prime truncation carries zero interior Ward residue in the nontrivial-zero region.

## 7 SRWC: the final soft-proof closure principle

**SRWC - Soft Renormalized Ward Continuity.** Let the finite exact linear-prime systems be completed by the canonical Hilbert-Schmidt/det<sub>2</sub> arithmetic completion described above. The cyclic/Ward residue class is continuous under this canonical completion. In particular, an interior residue that vanishes for every finite prime cutoff cannot be created as a new quantized residue in the completed renormalized connection.

Symbolically, if  $\mathfrak{R}_X(\Gamma)$  denotes the finite Ward residue on a contour  $\Gamma \Subset \Omega$  and  $\mathfrak{R}_{\text{ren}}(\Gamma)$  the completed residue, SRWC asserts

$$\mathfrak{R}_X(\Gamma) = 0 \quad \forall X \quad \implies \quad \boxed{\mathfrak{R}_{\text{ren}}(\Gamma) = 0}.$$

The completed connection is identified with

$$\boxed{\omega_{\text{ren}} = d \log F_{\text{lin}}}$$

on the nontrivial-zero domain. Therefore SRWC gives

$$\boxed{\oint_{\Gamma} d \log F_{\text{lin}} = 0}$$

for every sufficiently small contour around a putative interior zero.

### 7.1 Why SRWC is the right soft principle

SRWC is deliberately stronger than mere  $\mathcal{S}_2$  convergence of operators and deliberately weaker than assuming RH. Ordinary trace is not continuous on  $\mathcal{S}_2$ , so one may not argue

$$K_s^{(X)} \rightarrow K_s \text{ in } \mathcal{S}_2 \quad \implies \quad \text{Tr } K_s^{(X)} \rightarrow \text{Tr } K_s.$$

The canonical object is instead the quadratic determinant-line completion. SRWC says that the relevant Ward/Chern residue is compatible with that canonical completion and cannot acquire an integer defect absent at every finite stage.

**Hard-proof warning.** This is exactly the statement that must be unpacked in the hard-proof stage. The hard audit must specify the determinant-line topology, prove continuity of the residue/Chern class, establish the identification with  $d\log(V\zeta D_2)$  without circular analytic continuation, and control all domain and normalization issues.

## 8 Argument-principle closure

Assume  $F_{\text{lin}}$  has a zero  $\rho \in \Omega$  of multiplicity  $m \geq 1$ . Since zeros of a nonzero holomorphic function are isolated, take a small positively oriented contour  $\Gamma$  enclosing  $\rho$  and no other zero, with  $s = 1$  outside  $\Gamma$ . The argument principle gives

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{F'_{\text{lin}}(s)}{F_{\text{lin}}(s)} ds = m.$$

But SRWC gives

$$\frac{1}{2\pi i} \oint_{\Gamma} d\log F_{\text{lin}} = 0.$$

Hence  $m = 0$ , contradiction. Therefore

$$F_{\text{lin}}(s) \neq 0 \quad (\Re s > 1/2).$$

Since  $D_2$  is zero-free and  $V\zeta$  has its pole at  $s = 1$  removed by  $V$ ,

$$\zeta(s) \neq 0 \quad (\Re s > 1/2) \text{ for nontrivial zeros.}$$

The functional equation of the completed zeta function maps a nontrivial zero  $\rho$  to  $1 - \rho$ . A zero with  $\Re \rho < 1/2$  would therefore produce a zero with  $\Re(1 - \rho) > 1/2$ , impossible. Hence every nontrivial zero obeys

$$\Re \rho = \frac{1}{2}.$$

This is the Riemann Hypothesis.

## 9 Soft Proof v1.0 theorem

**Soft Proof v1.0 - Final Closure.** Accept the frozen exact NCHG/history identities, the explicit arithmetic Mellin/Poisson formulas, the Burnol Hardy endpoint used throughout the project, the canonical zero-free 2-regularized prime-power completion  $D_2$ , and SRWC as the final soft-layer structural principle. Then the Riemann Hypothesis follows.

The complete soft chain is

$$\begin{aligned}
H &= d_{\text{hist}} \mathbf{b}_H = \{q_r, b_r\}|_V, & C_{\text{tr}} &= R - P_\Lambda, \\
D_2(s) &= \det_2(I - K_s) \text{ is holomorphic and zero-free on } \Re s > \frac{1}{2}, \\
F_{\text{lin}} &= V\zeta D_2, & \text{Div}_\Omega(F_{\text{lin}}) &= \text{Div}_\Omega(\zeta), \\
C_{\text{lin}} &= R - P_1 \text{ is source-}Q\text{-exact,} \\
&\text{finite prime Ward residue} = 0, \\
&\text{SRWC} \implies \text{Res}_\Omega d \log F_{\text{lin}} = 0, \\
&F_{\text{lin}} \text{ zero-free on } \Omega, \\
&\zeta \text{ has no nontrivial zero with } \Re s > \frac{1}{2}, \\
&\xi(s) = \xi(1-s) \implies \boxed{\Re \rho = \frac{1}{2}}.
\end{aligned}$$

## 10 Equivalent hard-proof targets generated by SRWC

SRWC is the soft closure. For the hard proof, it should be replaced by one of the following explicit theorems.

### 10.1 Contour-continuity formulation

Construct canonical renormalized forms  $\omega_X$  and  $\omega_{\text{ren}}$  such that for every contour  $\Gamma \in \Omega \setminus \{1\}$ ,

$$\boxed{\lim_{X \rightarrow \infty} \oint_\Gamma \omega_X = \oint_\Gamma \omega_{\text{ren}}},$$

with

$$\omega_{\text{ren}} = d \log F_{\text{lin}}.$$

Since the left-hand integrals vanish, the right-hand integral vanishes.

### 10.2 Hurwitz formulation

Construct canonical zero-free finite renormalized determinant sections  $F_X$  such that

$$F_X \rightarrow F_{\text{lin}}$$

locally uniformly on  $\Omega$ . Hurwitz's theorem then implies that  $F_{\text{lin}}$  is either zero-free or identically zero. The Euler region excludes the latter, so  $F_{\text{lin}}$  is zero-free.

### 10.3 Determinant-line Chern formulation

Construct the determinant line of the  $S_2$  family and prove that the renormalized first Chern/Ward current has no interior delta mass beyond those present at finite rank. Since each finite truncation has zero interior divisor,

$$\boxed{c_1^{\text{ren}} = 0 \text{ on } \Omega}$$

and therefore the  $F_{\text{lin}}$  divisor vanishes.

## 11 Hard-proof prosecutor: exactly where the miyav returns

The soft proof is closed by convention; the hard proof must reopen the following questions.

- HP1. Topology.** Specify the exact  $S_2$ /graph/determinant-line topology in which the finite prime systems converge.
- HP2. Residue continuity.** Prove that the contour/Chern/Ward residue is continuous in that topology. Ordinary trace continuity on  $S_2$  is false and cannot be used.
- HP3. Canonical renormalization.** Show that the counterterm/regularization is uniquely fixed by the arithmetic and NCHG structures rather than tuned to remove zeros.
- HP4. Analytic identification.** Prove that the completed renormalized connection is  $d \log(V\zeta D_2)$  on  $\Omega$  without assuming its divisor behavior.
- HP5. Domain stability.** Control the Mellin/Poisson domains, the  $s = 1$  normalization, and contour deformation uniformly in the completion.
- HP6. Prime cutoff independence.** Prove that alternate admissible cutoffs/ordering prescriptions give the same residue class.
- HP7. No hidden zero data.** Ensure that no map, topology, counterterm, contour rule, or lift is defined using nontrivial zeta zeros.

**Hard-proof calibration: if any HP item fails, Soft Proof v1.0 remains a coherent project-internal soft proof but must not be promoted to an unconditional mathematical proof. If all HP items are proved, the remaining RH implication is already closed by the argument-principle chain above.**

## 12 False-positive and false-negative guards

### 12.1 False-positive prosecutor

The following moves are explicitly forbidden in any hard-proof upgrade:

- replacing SRWC by the statement “there are no residues” without deriving it;
- using the continuity of ordinary trace on  $S_2$ ;
- claiming that finite-rank zero-free functions automatically converge locally uniformly to  $F_{\text{lin}}$ ;
- defining a counterterm from known zero ordinates or from the Blaschke factor  $B$ ;
- using analytic continuation of an Euler product as if it automatically preserved a finite-cutoff Ward identity;
- erasing exact-sector determinant torsion without a determinant-line proof;
- confusing the zero-free  $D_2$  prime-power factor with a proof that the linear-prime factor is zero-free.

## 12.2 False-negative defender

The following failures do not automatically kill the research route:

- one cutoff scheme may fail while another canonical restricted-product cutoff succeeds;
- ordinary trace may diverge while  $\det_2$  and a determinant-line residue remain well-defined;
- a strict dg chain map may be replaced by a homotopy/derived map if the state-level normal class is controlled;
- a Hilbert-space proof may fail while a Frechet/distributional determinant-line proof survives;
- the  $p = 2$  or archimedean normalization may require a separate local counterterm, provided it is fixed independently and contributes no nontrivial-zero divisor.

## 13 Proof certificate

| ID      | Claim                               | Status                     | Evidence / formula                                                                                     |
|---------|-------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------|
| S100-01 | Ray differential                    | VERIFIED                   | $q_r^2 = 0$ .                                                                                          |
| S100-02 | Scalar transverse exactness         | VERIFIED                   | $\{Q_\delta, h_r\} = \delta I$ .                                                                       |
| S100-03 | Cartan transverse exactness         | VERIFIED                   | $\{Q_\delta, b_r\} _V = \delta H$ .                                                                    |
| S100-04 | History boundary                    | VERIFIED                   | $d_{\text{hist}} \mathbf{b}_H = H$ .                                                                   |
| S100-05 | Arithmetic vertical connection      | PROVED CORE                | ON $\nabla_{\text{ar}} = L + R - P_\Lambda, \partial_s \mathcal{E} = \mathcal{E} \nabla_{\text{ar}}$ . |
| S100-06 | Normal anomaly                      | EXPLICIT                   | $\mathfrak{A} = P_{K_B} \mathcal{E} C_{\text{tr}}$ .                                                   |
| S100-07 | Bad-zero anomaly                    | PROVED CORE                | ON A multiplicity- $m$ zero produces an $m - 1$ transverse jet.                                        |
| S100-08 | Prime-ray Hilbert-Schmidt threshold | VERIFIED                   | $K_s \in \mathcal{S}_2$ for $\Re s > 1/2$ .                                                            |
| S100-09 | $D_2$ convergence                   | PROVED HERE / ELEMENTARY   | $D_2 = \prod_p (1 - p^{-s}) e^{p^{-s}}$ locally uniformly zero-free on $\Omega$ .                      |
| S100-10 | Linear-prime divisor reduction      | PROVED HERE                | $F_{\text{lin}} = V \zeta D_2$ and $\text{Div}_\Omega(F_{\text{lin}}) = \text{Div}_\Omega(\zeta)$ .    |
| S100-11 | Prime-power safe sector             | PROVED HERE                | $C_{\geq 2} \leftrightarrow -D'_2/D_2$ absolutely convergent on $\Omega$ .                             |
| S100-12 | Linear-prime source exactness       | VERIFIED                   | $I \otimes C_{\text{lin}} = \{q_r \otimes C_{\text{lin}}, h_r \otimes I\}$ .                           |
| S100-13 | Finite cutoff residue               | VERIFIED                   | $\oint \omega_X = 0$ away from $0, 1$ .                                                                |
| S100-14 | SRWC                                | SOFT CLOSURE PRINCIPLE     | Canonical $\mathcal{S}_2/\det_2$ completion does not create a new quantized Ward residue.              |
| S100-15 | Argument-principle endpoint         | PROVED CONDITIONAL ON SRWC | ON $\text{SRWC} \Rightarrow F_{\text{lin}}$ zero-free on $\Omega$ .                                    |
| S100-16 | Functional-equation closure         | STANDARD                   | No zeros right of $1/2$ plus $\rho \leftrightarrow 1 - \rho$ gives RH.                                 |
| S100-17 | Soft Proof v1.0                     | CLOSED                     | Project-internal soft proof with SRWC reserved for hard audit.                                         |

| ID      | Claim                        | Status      | Evidence / formula                         |
|---------|------------------------------|-------------|--------------------------------------------|
| S100-18 | Standard unconditional proof | NOT CLAIMED | YET Requires hard proof of SRWC / HP1-HP7. |

## 14 Research-history relation

Soft Proof v1.0 should be read as the culmination of the following sequence of reductions:

1. the seed algebra produced the split  $H, S, J$  geometry and the exact time-ray complex;
2. history resolution showed the same transverse  $H$  direction is a boundary;
3. the arithmetic connection exposed the Mellin derivative as log-time plus pole and prime-power corrections;
4. the Burnol Hardy formulation reduced RH to a normal inner-factor obstruction;
5. v0.99 reduced the global completion problem to a single exact-state normal anomaly;
6. the final decomposition absorbed every  $k \geq 2$  prime-power contribution into the canonical zero-free  $D_2$  determinant;
7. the final soft obstruction became a single linear-prime renormalized Ward residue;
8. SRWC closes that residue at the soft layer; the hard layer must now justify the completion theorem itself.

## 15 Final freeze

**FINAL CHECKPOINT - SOFT PROOF v1.0 CLOSED.** In the project's terminology, the proof mechanism and all RH-specific algebraic/analytic implications have been reduced to a single general renormalized Ward-continuity principle, SRWC, whose role and possible failure modes are explicit. The next phase is not further soft-proof construction; it is a hard-proof audit of SRWC and the determinant-line completion.

The final soft-proof line is

finite exact prime Ward systems  $\xrightarrow{\text{det}_2 \text{ completion} + \text{SRWC}}$  no linear-prime interior residue  $\xrightarrow{\text{argument p}}$

## A Explicit $D_2$ logarithmic derivative

For  $\Re s > 1/2$ ,

$$\log D_2(s) = \sum_p (\log(1 - p^{-s}) + p^{-s}) = - \sum_p \sum_{k \geq 2} \frac{p^{-ks}}{k}.$$

Termwise differentiation is locally uniform on compact subsets of  $\Omega$ , yielding

$$\frac{D'_2}{D_2}(s) = \sum_p \sum_{k \geq 2} (\log p) p^{-ks}.$$

Thus the entire  $k \geq 2$  logarithmic derivative is holomorphic on  $\Omega$ .

## B Explicit finite-prime determinant section

In the Euler region define

$$F_X(s) = V(s) \exp \left( \sum_{p \leq X} p^{-s} \right).$$

Then

$$\frac{F'_X}{F_X} = \frac{V'}{V} - \sum_{p \leq X} (\log p) p^{-s} = \omega_X/ds.$$

For every contour  $\Gamma \Subset \Omega \setminus \{1\}$ ,

$$\frac{1}{2\pi i} \oint_{\Gamma} \frac{F'_X}{F_X} ds = 0$$

because  $F_X$  is zero-free there. The hard-proof Hurwitz route would establish locally uniform convergence of a canonical renormalized version of these zero-free sections to  $F_{\text{lin}}$ .

## C References and provenance

1. Misteria and GPT-5.6 Sol, *RH-GHM-NCHG Research Ledger*, 12 August 2026.
2. Misteria and GPT-5.6 Sol, *Supplemental I: C1 Update, C1 Faithfulness and C2 Global Factorization Specification*, 12 August 2026.
3. Misteria and GPT-5.6 Sol, *C2 Ledger*, 12 August 2026.
4. Misteria and GPT-5.6 Sol, *Naive Conditional Proof: Global Relative Verticality Version*, 12 August 2026.
5. Misteria and GPT-5.6 Sol, *Soft Proof v0.97*, 12 August 2026.
6. Misteria and GPT-5.6 Sol, *Soft Proof v0.99: Semi-Unconditional Consistency-Closure Version*, 12 August 2026.
7. Misteria and GPT-5.6 Sol, *NCHG v0.2: Nonassociative Causal-Holonomy Gravity*, 2026.
8. Misteria and GPT-5.6 Sol, *QMS Theory v1.0*, 2026.
9. J.-F. Burnol, *An adelic causality problem related to abelian L-functions*, Journal of Number Theory 87 (2001), 253–269; arXiv:math/0001013.
10. J.-F. Burnol, works on the local conductor operator and explicit formula, 1998–1999.
11. R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277, revised 2013.